

```

1 ROWS (1+ROUND (1, 1))
1 ROWS (1+ROUND (ROUND (PI () , 2) , ROWS (SIN (3))))
4. 1 1+ROUND (ROUND (PI () , 2) , ROWS (SIN (3)))
2. 070796327 AVERAGE (PI () , ROUND (TRUE () , PI ()))
3. 141592654 MAX (MAX (PI ()))
3 COUNT (AVERAGE (pi () ) , PI () , 1+PI ())
3 ROUND (ROWS (ROUND (PI () , 1+PI ())) , 1+PI ())+ROWS (ROWS (PI () +1))+1*ROUND (1, 1)
3. 141592654 MIN (PI ())
3. 141592654 MDETERM (MIN (PI ()))
1 COLUMNS (PI ())
$1. 05 NPV (2, PI ())
0 SUMIF (A1, PI ())
1 COLUMNS (IF (C13, SUMIF (C13, PI ()), SIN (1)))
1 COLUMNS (SUMIF (A1, PI ()))
1 COLUMNS (IF (SIN (0) , SIN (1) , SIN (2)))
1 MDETERM (ABS (A1))
1 ROWS (ROUND (INDEX (C1, 1, ROUND (1, 1)) , PI ()))
1 INDEX (A1 :B2, 1, 1)
1 ROUND (1, 1)
1 MID (1, 1, 1)
2 1+1
1 COLUMNS (ROUND (A1, PI () +SIN (1)))
1 COLUMNS (ROUND (INDEX (A1, 1) , PI ()))
55 OFFSET (A1, 2, 2)
66 OFFSET (INDEX (A1 :B2, 2, 2) , 2, 2)
#NUM! IRR (PI () , 0)
1 COLUMNS (INDEX (A1 :B2, 2, 2))
#NUM! (1+IRR (SUMIF (D :D, ROW (INDIRECT (MIN (D :D)&" : "&MAX (D :D))) , E :E) , 0)) ^365-1
#DIV/0! MIRR (PI () , 0. 01, 0. 02)
#N/A VLOOKUP (1, PI () , 1)
0 SUMPRODUCT ((' Sheet1' !C3:C751=Sheet1!F22)*(' Sheet1' !D3:D751=Sheet1!G22) , 'S
0 COUNT (C1 :C2)
1 COUNT (C1*1)
4 MODE ({2. 0, 3. 0, 4. 0;6. 0, 5. 0, 4. 0})
12 MMULT (I33 :I34, J33 :K33)
3 COLUMNS (SIN ({2, 3, 4}+1)+1)
4 ABS (FREQUENCY ($E$32 :$E$38, $E$32 :$E$38))
0. 833333333 AVERAGE (FREQUENCY ($E$32 :$E$38, $E$32 :$E$38))
8 COUNT (ABS (FREQUENCY ($A1 :$A5, $A1 :$A5)))
0 MODE (ABS (FREQUENCY ($E$32 :$E$38, $E$32 :$E$38)))
5 ABS (FREQUENCY ($E$32 :$E$38, $E$32 :$E$38)+1)
0 IF (C41, D41, E41)

```

Failing cases:

```

1 AVERAGE (ABS (IF ((FREQUENCY ($E$32:$E$38, $E$32:$E$38)-1)>0, 1)))
1. 875 AVERAGE (ABS (IF ((FREQUENCY ($E$32:$E$38, $E$32:$E$38)-1)>0, 1)+FREQUENCY ($E$3
2 COLUMNS (IF (C53, E49:F49, {5, 6, 7, 8}))
3 COLUMNS (IF (C53, {3, 4, 5}, {6, 7}))

```

Excel encodes PI() as 'array'. Not sure what rule justifies this. POI encodes it as
3. 141592654 INDEX (PI (), 1)

These formulas involve constant arrays (which POI 3.1 does not parse yet)

```

0. 816496581 STDEV (2, ROUND ({1. 0;2. 0;3. 0}, 1))
22. 14159265 INDEX (MMULT ({1. 0, 2. 0;3. 0, 4. 0}, {5. 0, 6. 0;7. 0, 8. 0})+PI (), K5, K6)
19 TRANSPOSE (MMULT ({1. 0, 2. 0;3. 0, 4. 0}, {5. 0, 6. 0;7. 0, 8. 0}))
1 SUM (MMULT ({1. 0}, {1. 0}))
134 SUM (MMULT ({1. 0, 2. 0;3. 0, 4. 0}, {5. 0, 6. 0;7. 0, 8. 0}))

```

Something wierd going on with the TRANSPOSE function

```

0 TRANSPOSE (INDEX (C1, 1, 1))
3. 141592654 TRANSPOSE (PI ())

```

```

SUM (VLOOKUP ({ "a", "b", "b"}, C72:D75, 2))

```

#N/A

```

0. 25 AVERAGE (IF (FREQUENCY ($E$32:$E$38, $E$32:$E$38)>0, 1, 0))
1 AVERAGE (IF ((FREQUENCY ($E$32:$E$38, $E$32:$E$38)-1)>0, 1))
40 MMULT (IF (C35, I34:J34, K34:K36), IF (C35, K34:K35, I35:K35))
59 MMULT (IF ($I$36, {2, 3}, {4, 5}), IF ($I$37, {6;7}, SIN(8)))
4 INDEX (INDEX ({1, 2;3, 4}, {2, 1}, 2), 1, 1)

```

HEET1' !AF3:AF751)

2	1
2	0
2	0
2	0
4	1
4	0
4	0
	0
	0.25

2 a
3 b

3			
4	3	8	9
5	3	3	9
0	7	3	9
1			
	0	1	1
	59		
a	b		
	4	2	

2:\$E\$38,\$E\$32:\$E\$38)))

1

'value' which Excel seems to tolerate

a	5
b	4
c	2
d	1